SAMPLE

Name: _____ Section: _____

One page of handwritten notes is permitted: one side of one standard 8.5 x 11-inch sheet.

Write all answers by hand in Python. Use clear indentation. Assume all amounts are nonnegative and each list is nonempty. Given variables are already defined; your code should work beyond the sample data.

Use this sample to practice writing code without running it first.

Question 1 | Customer order and shipping | 30 points

An online retailer applies a discount first, then determines shipping from the discounted order total.

Order amount before discount	Discount
Below $200	0%
$200 or more	10%

After the discount, use the discounted order total to determine shipping:

Discounted order total	Shipping charge
Below $50	$10
$50 to less than $100	$5
$100 or more	$0

```
order_total = 225
```

Use conditionals to calculate the discount, discounted order total, shipping charge, and final amount owed. Print the discounted order total, shipping charge, and final amount owed.

Write your code below.

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Question 2 | Customer satisfaction analysis | 35 points

A company collected the following customer satisfaction scores. Each list value is one customer score.

```
scores = [8, 5, 9, 6, 10, 4]
```

High score: 8 or more. Low score: 5 or less.

Use a loop to calculate the total of all scores, count high scores, count low scores, and count how many scores were processed. After the loop, calculate the average score. Print the total, average, high-score count, and low-score count.

Calculate the results using the loop. Do not use sum(), max(), min(), sorted(), or comprehensions. Do not hard-code the results.

Write your code below.

SAMPLE

Name: _____

Question 3 | Employee performance bonus processing | 35 points

Part A: 18 points | Part B: 14 points | Part C: 2 points | Organization: 1 point

A company assigns a fixed bonus based on an employee performance score.

Performance score	Bonus
Below 70	$0
70 to 84	$500
85 to 94	$1,000
95 or more	$1,500

Part A | Define the function | 18 points

`calculate_bonus(score)`

Write this function with one parameter. Use conditionals to determine the bonus amount and return that amount. Do not print inside the function.

Write your function below. Parts B and C are on page 4.

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Name: _____

Question 3 | Continued

Part B | Process all employees | 14 points

```
performance_scores = [82, 96, 68, 91, 75]
```

Outside the function, initialize a total and loop through this list. Call your Part A function for every score. Print each score and its bonus; accumulate all bonuses. Print the total bonus expense after the loop. Do not repeat the bonus rules here.

Write the calling code below.

Part C | Explain the design | 2 points

In 1-2 sentences, explain why return is useful in calculate_bonus() and how the returned value is used in Part B.

Additional answer space: label the question and part you are continuing.

Preparation reminder: practice writing complete solutions on paper before checking them in Python.